

UNIT – 2
ASSESSMENT – 3

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SOLUTION

1. Title: A* Search Algorithm

The A* Search Algorithm starts from the source node and expands the node with the lowest $f(n)$ value, where $f(n) = g(n) + h(n)$. The algorithm maintains an Open List containing nodes to be explored and a Closed List containing visited nodes. At each iteration, the node with the minimum $f(n)$ value is selected until the goal node is reached. This approach ensures that the shortest and optimal path is obtained efficiently.

2. Title: Minimax Algorithm with Alpha-Beta Pruning

The Minimax algorithm evaluates the game tree by recursively exploring all possible moves. MAX selects the maximum value, while MIN selects the minimum value. During evaluation, Alpha (α) represents the best value for MAX, and Beta (β) represents the best value for MIN. Whenever $\beta \leq \alpha$, the remaining branches are pruned because they cannot influence the final decision. This reduces computation while producing the same optimal result.